

Clock – Valid pattern generation Functional Design Specifications

UCIe 3.0 physical layer

Version 1

Block Owner

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2 Revision History

Version	Date	Author(s)	Revision Notes	Owner Approval

3 Overview

This module generates gated clock and valid output. It supports two operating modes — half-rate and quarter-rate — and produces selectable output patterns based on a 2-bit pattern type selector.

4 Operation and Description

4.1 Digital Interface

4.1.1 Parameters Names

Parameter Name	Default	Description

4.1.2 Ports Names

Port Name	Port Width	Port Type	Description
i_valid	1	Input	Data valid indicator
i_hclk	1	Input	Half-rate clock
i_halfrate	1	Input	Mode select: 1 = half-rate, 0 = quarter-rate
i_reset	1	Input	Synchronous reset for clock dividers and PWM
i_pattern_type	2	input	Selects the output pattern type
o_clk_p	1	Output	Gated primary output clock
o_clk_n	1	Output	Gated complementary / quadrature output clock
o_valid	1	Output	Gated valid output
o_track	1	Output	Mirrors o_clk_p

4.2 Functional Description

1. Clock Generation

- Half-Rate Mode (i_halfrate = 1):
 - i_hclk is used directly as the primary clock. Its complement drives o_clk_n to produce a 180° phase-shifted pair.
- Quarter-Rate Mode (i_halfrate = 0):
 - Two clock_divider instances divide i_hclk and ~i_hclk respectively, producing two quadrature clocks (w_qclk_1, w_qclk_2) with a 90° phase offset between them. These drive o_clk_p and o_clk_n independently.

2. Duty Cycle Shaping (PWM)

- Three independent PWM counters run on their respective clocks (i_hclk, w_qclk_1, w_qclk_2). Each counter cycles from **0 to 23** (24 periods total), asserting its enable signal high for the first **16 cycles** and low for the remaining **8**. This produces a precise **2/3 high, 1/3 low duty cycle** (~66.7%), modeling the effect of a ÷24 clock with DLL-adjusted duty cycle. This is a behavioral PWM approximation.

3. Pattern Type Encoding

i_pattern_type	Clock Output	Valid Output	Use Case
2'b00	Suppressed	Enabled	Valid-only pattern
2'b01	Enabled (PWM-gated)	Suppressed	Clock-only pattern
2'b10	Enabled (always on)	Enabled	Active/data pattern
2'b11	Enabled (always on)	Suppressed	Active with no data

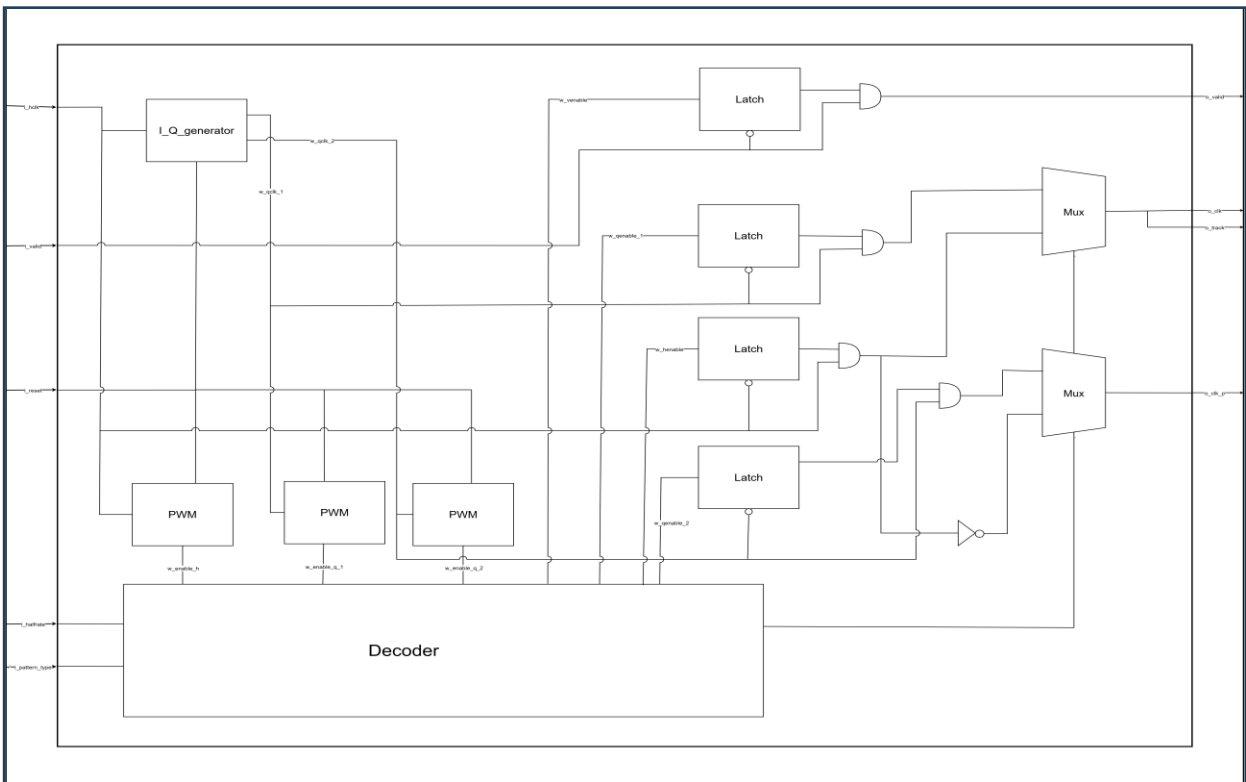
4. Clock Gating (ICG Pattern)

- The module implements latch-based Integrated Clock Gating (ICG). Enable signals (w_henable, w_qenable_1, w_qenable_2, w_venable) are updated only when their respective clock is LOW, which is the standard glitch-free safe update window. When the clock is HIGH, the enables hold their previously latched values. The final gated clock output is produced by ANDing the clock with its enable.
- One key behavioral detail: w_venable is evaluated only when i_valid is LOW. This means the valid enable latches its value during the de-asserted phase of i_valid, holding it stable while valid is high — mirroring the same ICG philosophy applied to the clock path.

5. Track Signal

- o_track is a direct wire assignment mirroring o_clk_p, as required by the UCle specification for receiver clock tracking.

4.3 Block Diagram



4.4 Timing Diagram

